COOLING ENVIRONMENTS

OCP Air versus Liquid –
Where Does Air Cooling Still Win for General Compute

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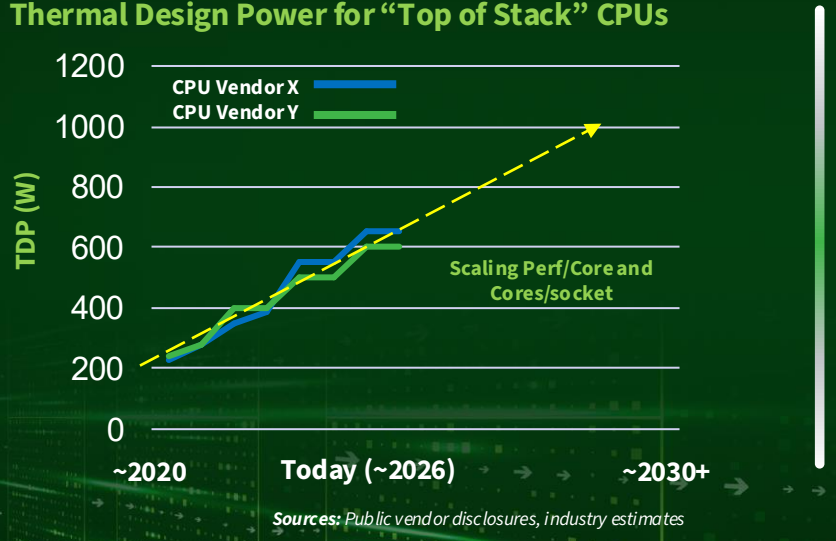
Agenda

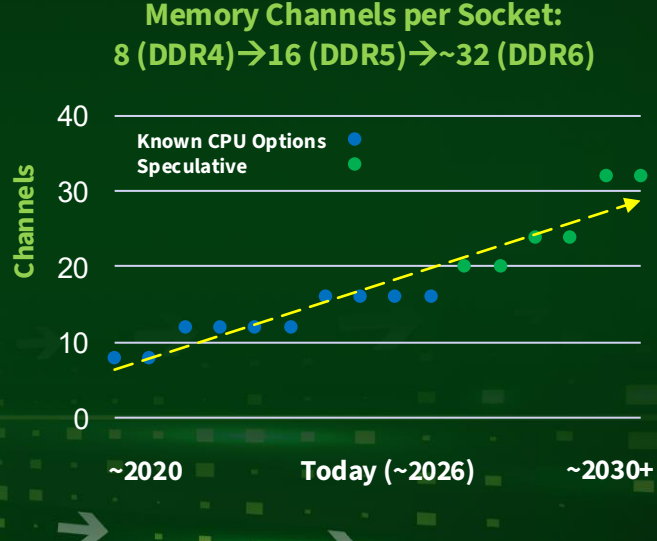
- Industry Trends
- Cooling Options
- Benefits of Wider Racks
- Trade-offs and Next Steps



Industry Trend: Higher Silicon Power & More Memory Channels

- CPU TDP already beyond 650W in high-end server-class roadmaps and trending towards 1000W+
- DDR6 memory capacity/bandwidth for 2S platforms may require up to 32 channels per socket including DIMM, VRM, and routing clearances





Sources: *Public vendor disclosures, industry estimates*

Platforms must adapt to higher TDPs, channel counts, and clearances in a cost-effective manner

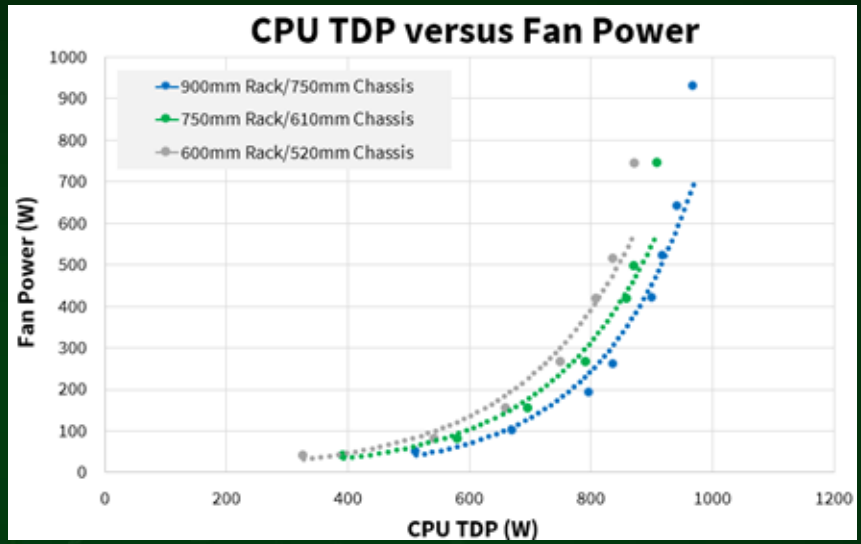
Available Paths

Paths	Pros / Cons
Taller Air Chassis	+ Minimal facility impact - Limited gains (airflow, heatsink volume, board area), lower rack density
Wider Air Chassis	+ More airflow area, larger heatsinks, DDR6 I/O placement headroom - Requires ecosystem alignment on rack form factor
Hybrid Liquid (CPUs and DIMMs only)	+ Highest thermal capability, cold plate design less complex - Higher infrastructure cost, operational complexity, and service requirements
Full Liquid (Fan-less)	+ Highest thermal capability - 50% higher liquid cooling infrastructure cost (compared to Hybrid Liquid), operational complexity, and service requirements

Focus: Wider Air Chassis path — cost and row-level density are the primary criteria

Wider Air Chassis Improves Fan Power Efficiency

- Considers from 600mm to 900mm wide racks
- 900mm width increases airflow area versus 600mm by ~50%
- At equal flow, velocity drops roughly one-third; pressure-drop scales approximately with velocity^2
- ~20-30% less fan power per kW cooled with wider chassis
- Lower impedance allows fans to operate at lower RPMs for the same thermal load
- ORW dimension is overkill for general compute

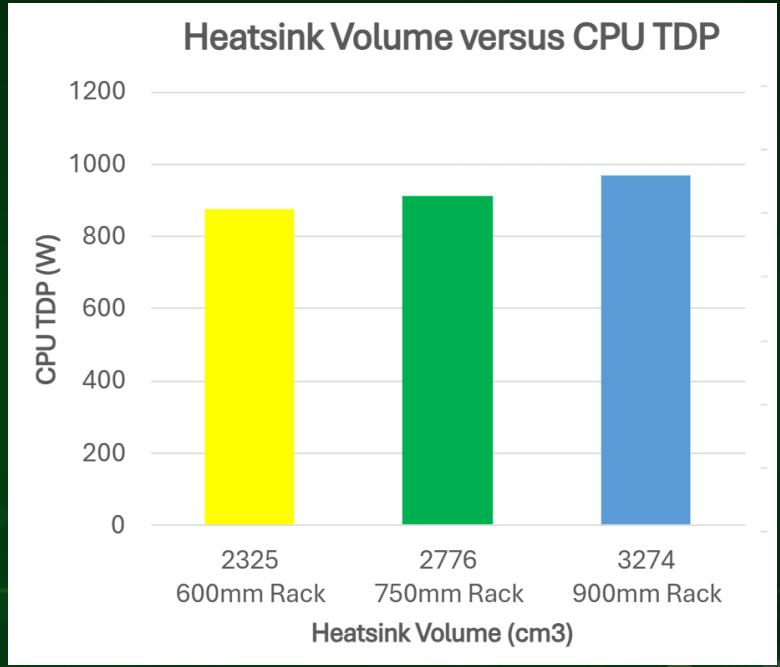


Note: *Fan power and TDP references 1.5OU system*

Wider Air Chassis shifts fan-power curve down at high TDP;
Wider Air Rack increases airflow lowering fan power by ~20-30%

Available Heatsink Volume by Rack Width

- Target to support CPU TDP from 650W to 1000W+ with measurable thermal margin
- 900mm chassis width allows larger heatsinks with volumetric increases to fin area/depth and duct cross-section
- More surface area improves heat transfers and wider flow paths minimizing excessive pressure drop

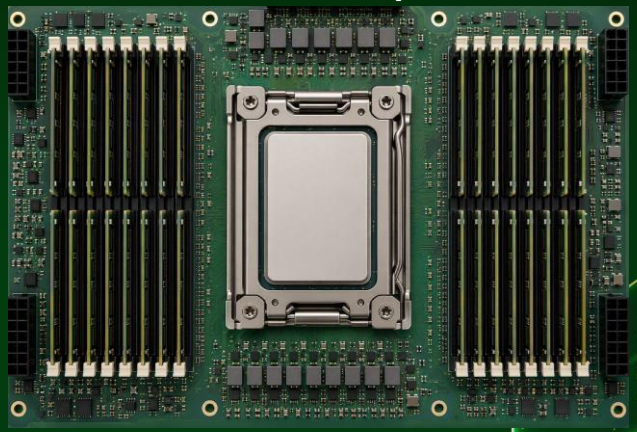


Larger heatsink with more fin area avoids excessive pressure drop without high impedance

DDR6-Era Systems Have Wider Footprint

- Memory roadmap increasing channels per socket from 16 (DDR5) → 20 → 24 → 32 (DDR6) — platforms have to be laid out for where that curve ends, not where it starts
- 32 channel 2S board needs DIMM spacing, VRM clearance, routing corridors, and serviceable keep-outs — a general-compute layout problem, not a single point design
- Added 300mm width allows 2S CPU spread layouts reducing CPU/DIMM/VRM thermal shadowing and allowing room for >50% more I/O or disk while preserving airflow paths
- DDR6 board redesign is the lowest disruption moment to adopt the wider envelope — thermal and layout headroom captured in one step

Visual Concept



Estimated ~300mm
32CH DDR6 example

Memory channel growth makes width a board-layout constraint, not just a thermal constraint

Trade-off Matrix

Factors	Taller Air Chassis	Wider Air Chassis	Hybrid Liquid	Full Liquid
Thermal	650 to 1000W+ pending rack test	650 to 1000W+ pending rack test	Highest heat flux / no air margin	Highest heat flux / no air margin
Chassis	Less rack density, limited width support for DDR6	Maintain rack density, more fans, bigger heatsinks	Coldplate design less complex	Fan-less design
Facility	More racks to maintain Compute density	Rack-envelope change	CDU + coolant distribution	CDU + coolant distribution
Energy	Baseline	~20–30% lower fan W/kW	Lower fan power + pump/CDU load	No fan power + pump/CDU load
Operations	Standard FRUs and service	Standard FRUs and service	Fluid, leak, corrosion controls	Fluid, leak, corrosion controls

Wider Air-Cooled Racks are the lowest-friction path for general compute;
Liquid makes sense only above the air envelope

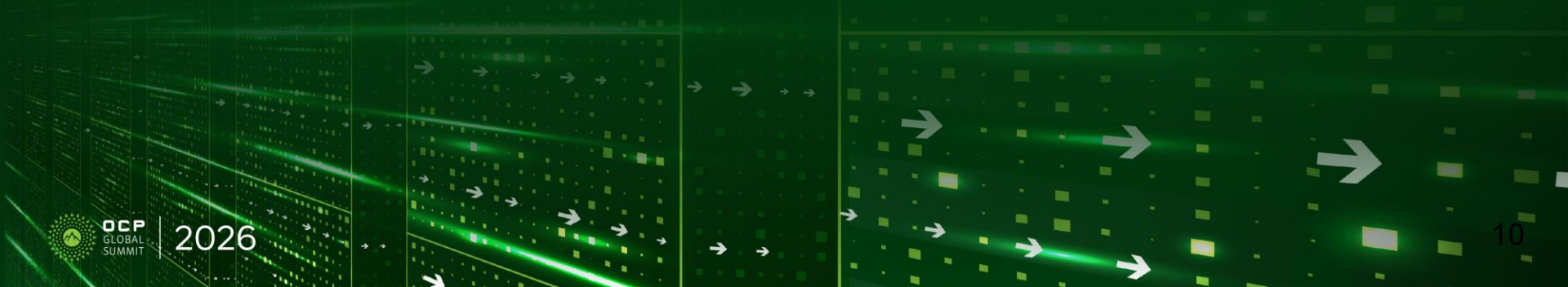
Conclusion

- Wider chassis improves airflow area by ~50% reducing velocity and drives ~20–30% fan power savings per kW cooled
- Wider chassis enables larger heatsink envelopes extending air cooling into the 1000W+ CPU TDP range
- Wider chassis enables placement of 2S CPU side by side with DDR6 layouts with up to 32 memory channels per socket
- Wider rack extends the useful life of air cooling for general compute by improving airflow impedance, heatsink volume, and board layout headroom

Liquid remains the right answer when measured thermal margin, heat flux, or site constraints exceed the wider air envelope

Call to Action

- What this is: Early data on the 900mm air-cooled envelope, shared for community discussion — not a contribution, spec, or new workstream
- What we're asking: Does this match your measurements and where does the air envelope break first for your workloads?
- Where to continue it: Share counter data and use cases to the existing Rack & Power and Cooling Environments community calls
- How to reach us: Find us after this session — happy to walk through test methodology and rack-level results




Thank
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